CSE 29
Lecture 6 Summary

January 22, 2026

📣 Logistical Things

- Assignment 2 is due next Thursday (1/29), which includes:
  - PSet 2 on Prairie Learn
  - PA 2
  - Design Questions 2
  - We have gone over all the things you need to complete PA2
- Assignment 1 resubmission is also due next Thursday (1/29)
  - More details on resubmission policy: https://ucsd-cse29.github.io/wi26/#assignments

🧠 Review Questions

Answers in speaker notes!

For the string “Jéan” answer the following questions:

Q1: What is the UTF-8 encoding?

Q2: What is the UTF-32 encoding?

Q3: What is `strlen` of the UTF-8 encoding? The UTF-32?

J = 74 = 0x4A

é = 233 = 0xE9

a = 97 = 0x61

n = 110 = 0x6E

Answer 1: 0x4A 0xC3 0XA9 0X6E

strlen returns 5

Answer 2: 0x00 0x00 0x00 0x4A 0x00 0x00 0x00 0xE9 0x00 0x00 0x00 0x61 0x00 0x00 0x00 0x6E

strlen returns 0 for this version (strlen is NOT compatible with UTF-32 encoding!)

Another version of UTF-32 depending on byte ordering: 0x4A 0x00 0x00 0x00 0xE9 0x00 0x00 0x00 0x61 0x00 0x00 0x00 0x6E 0x00 0x00 0x00
https://en.wikipedia.org/wiki/UTF-32

strlen returns 1 for this version

Pointers

`string.h`

`int strlen(char *s)` class notation would be `int strlen(char s[])`

`void strcpy(char *dest, char *src)` Copies from src to dest

`void strcat(char *dest, char* src)` Appends src to end of dest

`char str[]` and `char *str` <u>mean the same thing</u> in function arguments

`char*`: an address in memory where we can access `char` data

`T*`: an address in memory where we can access T data

`T` could be `int, double, int16_t, char* etc`

Pointer Types

address = number representing a place in memory (storage in our computer)

For a pointer `T *p`:

`p[index]` looks up the value offset by `index` from `p` in memory

`p[index] = v` change the value offset by `index` from `p` to the value `v`

If `x` is a variable of type `T`

`&x` => evaluates to a `T*` that is the **address** where `x` is stored

Questions

- Could you return pointers?
  - Yes! This is one of the things that differentiates `char[]` and `char*`.
- Are pointers also stored at an address?
  - Yes, pointers themselves also live somewhere in memory so they have addresses.

Look through this code, what observations do you have?

```c
#include <stdio.h>
#include <stdint.h>

void wheresmystuff(char* s) {
  int x = 12;
  uint8_t y = 53;
  char z = 9;
  double ns[] = { 4.0, 3.0, 9.0 };

  printf("x=%d, %lu bytes, starts at: %p\n", x, sizeof(x), &x);
  printf("y=%hhu, %lu bytes, starts at: %p\n", y, sizeof(y), &y);
  printf("z=%hhd, %lu bytes, starts at: %p\n", z, sizeof(z), &z);
  printf("ns=[%f,%f,%f], %lu bytes, starts at: %p\n",
         ns[0], ns[1], ns[2], sizeof(ns), &ns);

  printf("s=\"%s\"@%p, %lu bytes, starts at: %p\n", s, s,
      sizeof(s), &s);
}

int main() {
  char str[] = "14 char string";
  wheresmystuff(str);
  printf("\nstr takes up %lu bytes starting at: %p\n",
      sizeof(str), &str);
}
```

Output

```
$ # NOTE - Joe ran this on his Macbook to get this output
$ gcc wheresmystuff.c -o wheresmystuff
$ ./wheresmystuff
x=12, 4 bytes, starts at: 0x16f4c6e44
y=53, 1 bytes, starts at: 0x16f4c6e43
z=9, 1 bytes, starts at: 0x16f4c6e42
ns=[4.000000,3.000000,9.000000], 24 bytes, starts at:
    0x16f4c6e50
s="14 char string"@0x16f4c6e98, 8 bytes, starts at:
    0x16f4c6e48

str takes up 15 bytes starting at: 0x16f4c6e98
```

%p to print pointers

%f to print non-integer number (contrast from %d)

Observations

- Line 16
- x,y,z seem to be stored close together (base on the address)
- sizeof operator returns the size of the entire array and not the pointer for ns
  - 24 bytes for three 8-byte doubles
  - No this does not work
- s is a 15 character array, but its size is 8 bytes

Memory Diagram

Each row is 8 bytes of data

Convenient for mapping out memory and variable storage!

Variable/Role	Address	Data							
		0/8	1/9	2/A	3/B	4/C	5/D	6/E	7/F
	0x...00								
	0x...08								
	0x...10								
	0x...18								
	0x...20								
	0x...28								
	0x...30								
	0x...38								
	0x...40								
	0x...48								
	0x...50								
	0x...58								
	0x...60								
	0x...68								
	0x...70								
	0x...78								
	0x...80								
	0x...88								
	0x...90								
	0x...98								
	0x...A0								
	0x...A8								
	0x...B0								
	0x...B8								
	0x...C0								
	0x...C8								
	0x...D0								
	0x...D8								
	0x...E0								
	0x...E8								
	0x...F0								
	0x...F8								

Where are the variables stored?

```c
#include <stdio.h>
#include <stdint.h>

void wheresmystuff(char* s) {
  int x = 12;
  uint8_t y = 53;
  char z = 9;
  double ns[] = { 4.0, 3.0, 9.0 };

  printf("x=%d, %lu bytes, starts at: %p\n", x, sizeof(x), &x);
  printf("y=%hhu, %lu bytes, starts at: %p\n", y, sizeof(y), &y);
  printf("z=%hhd, %lu bytes, starts at: %p\n", z, sizeof(z), &z);
  printf("ns=[%f,%f,%f], %lu bytes, starts at: %p\n",
         ns[0], ns[1], ns[2], sizeof(ns), &ns);

  printf("s=\"%s\"@%p, %lu bytes, starts at: %p\n", s, s,
      sizeof(s), &s);
}

int main() {
  char str[] = "14 char string";
  wheresmystuff(str);
  printf("\nstr takes up %lu bytes starting at: %p\n",
      sizeof(str), &str);
}
```

```
$ # NOTE - Joe ran this on his Macbook to get this output
$ gcc wheresmystuff.c -o wheresmystuff
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x=12, 4 bytes, starts at: 0x16f4c6e44
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z=9, 1 bytes, starts at: 0x16f4c6e42
ns=[4.000000,3.000000,9.000000], 24 bytes, starts at:
    0x16f4c6e50
s="14 char string"@0x16f4c6e98, 8 bytes, starts at:
    0x16f4c6e48

str takes up 15 bytes starting at: 0x16f4c6e98
```

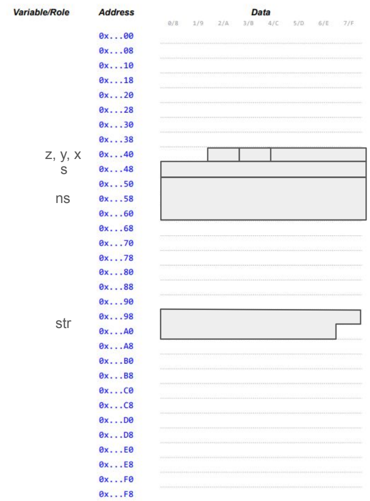

Where are the variables stored? (answer)

%p - pointers
%f - floating point

AH

```c
#include <stdio.h>
#include <stdint.h>

void wheresmystuff(char* s) {
  int x = 12;
  uint8_t y = 53;
  char z = 9;
  double ns[] = { 4.0, 3.0, 9.0 };

  printf("x=%d, %lu bytes, starts at: %p\n", x, sizeof(x), &x);
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  printf("ns=[%f,%f,%f], %lu bytes, starts at: %p\n",
         ns[0], ns[1], ns[2], sizeof(ns), &ns);

  printf("s=\"%s\"@%p, %lu bytes, starts at: %p\n", s, s,
      sizeof(s), &s);
}

int main() {
  char str[] = "14 char string";
  wheresmystuff(str);
  printf("\nstr takes up %lu bytes starting at: %p\n",
      sizeof(str), &str);
}
```

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str takes up 15 bytes starting at: 0x16f4c6e98
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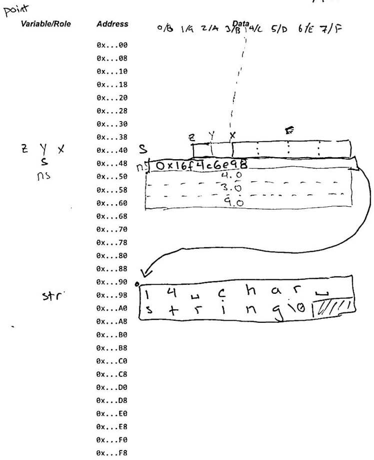

Questions

- How does the program decide where to put variables?
  - There is an algorithm in gcc to decide this. It would be different on different versions of gcc and computers
- sizeof is a compile time concept

Hashing

Examples of generated hashes

```
$ git commit -m “almost done”
[main 211935b] almost done
1 file changed…
$ git log
Commit 211935b0cf003 … (40 hex char = 20 bytes)
$ ssh jpolitz@ieng6
ED25519 key fingerprint SHA256: 8avDdt0
```

You have seen this in lab or doing the PA!

How are these numbers generated?

`SHA256(char *input, int len char* hash)`

More info in speaker notes 💡

- `input` is not a C-string and doesn't need to be null terminated
- `len` is the number of chars to process with the hashing function
- `hash` is an output parameter that is written over

Calculates a fixed-size (32 byte) "hash" of that input data.

- deterministic (NOT random)
- "one-way": hard to guess the input from output
- unpredictably distributed: similar inputs produce very different outputs

char *input is JUST a byte array, not a C string. Not every char* you see will be a C string.

Joe’s Notes (11am)

A1

For this string: Jéan (é is code point 233)

Q1: What is the UTF8 encoding?

Q2: What is the UTF32 encoding?

Q3: What is strlen of the UTF8 encoding? The UTF-32?

	J	é	a	n
	74	233	97	110
code points in hex:	0x4A	0xE9	0x61	0x6E

~~0b000~~
0b01001010

0b11110 1001

UTF32 encoding:

J				é			
0x00	0x00	0x00	0x4A	0x00	0x00	0x00	0xE9
0x4A	0x00	0x00	0x00	0xE9	0x00	0x00	0x00

strlen returns 0 for top version
1 for bottom version

UTF8: 0x4A $\overbrace{\text{0xC3 0xA9}}^{é}$ 0x61 0x6E

strlen returns 5

A2

$\boxed{\text{string.h}}$

```
int strlen(char* s)
void strcpy(char* dest, char* src)
void strcat(char* dest, char* src)
```

class notation would be int strlen(char s[])

copies bytes from src to dest

appends src to the end of dest

char str[] vs. char* str

In function arguments, these mean the same thing

char* : an address in memory where we can access char data

T* : an address in memory where we can access T data

T could be
int
double
int16_t
char*

Pointer Types

address = number representing a place in memory (storage in our computer)

For a pointer T* p;

p[index] looks up the value offset by index from p in memory

p[index] = v change the value offset by index from p to the value v

x is a variable of type T
&x ⇒ evaluates to a T* that is the address where x is stored

A3

Hashing

```
$ git commit -m "almost done"
[main 211935b] almost done
 1 file changed ...
$ git log
commit: 211935b0cf003 ...
```

(40 hex char = 20 bytes)

Open File

```
$ ssh jpolitz@ieng6
ED25519 key fingerprint SHA256: 8avDd+O...
```

just a byte array

32 bytes

SHA256(char* input, int size, char* hash)

calculates a fixed-size "hash" of that data

- deterministic
- "one-way": hard to guess the input from output
- unpredictably distributed: similar inputs produce different outputs

Joe’s Notes (12:30pm)

string.h | int strlen(char str[]) | A1

int strlen(char* str)
void strcpy(char * dest, char* src) copies src to dest
void strcat(char * dest, char* src) appends src to end of dest

~~char*~~

char* s vs. char s[]
These mean the same thing as function arguments

char* is an address where we can access char data
T* is an address where we can access T data

T could be
int16_t, double, float, int, uint64_t, char*

T* is a Pointer Type
pointer = address (of data in memory)
a number

T* ptr;

Operations:
ptr[index] look up data in memory starting at ptr and adding offset of index

ptr[index] = v change data in memory starting at ptr and adding offset of index

x is variable of type T
&x evaluates to a pointer (type T*) holding ~~holding~~ the address where x is stored

A3

Hashing

$ git commit -m "almost there!"
[main $\boxed{0b1a569}$] almost there!
1 file changed
commit hash
$git log
$\boxed{0b1a569a1490\cdots}$ [40 hex chars]
Author: Joe ...

$ ssh jpolitz@ieng6
The authenticity of host ieng6 cannot be established ...
.... key fingerprint is SHA256: $\boxed{8vaDd+0\cdots}$ (~40 char)
hash of a "key" on server

can ~~ein~~ be any length
not a C string
the # of chars in input

void SHA256(char* input, int len, char* hash)
32 bytes (output param)

- deterministic - same input produces same hash
- one-way - cannot compute the input given hash
- unpredictably distributed - similar inputs have very different hashes